

HDOC Day-32

Menu-Driven Matrix Operations Using Separate Functions

Aim

To write a menu-driven C program using separate functions to perform matrix addition and matrix multiplication on variable-length 2D arrays, and to demonstrate that these operations are not commutative where applicable.

Problem Statement

1. Get two variable-length 2D arrays and prove that matrix addition is commutative.
2. Get two variable-length 2D arrays and prove that matrix multiplication is not commutative.
3. Exit the program.

Algorithm

4. Start the program and display a menu with Addition, Multiplication, and Exit options.
5. Read the user choice and use a switch statement to call the appropriate function.
6. For matrix addition, read rows and columns, then input matrices A and B of the same order.
7. Calculate $A + B$ and $B + A$. Compare the two result matrices. If all corresponding elements are equal, addition is commutative.
8. For matrix multiplication, read dimensions of A and B such that both $A \times B$ and $B \times A$ are defined. A convenient demonstration uses two square matrices of the same order.
9. Calculate $A \times B$ and $B \times A$. Compare the products. If they differ, the chosen matrices demonstrate that matrix multiplication is not commutative.
10. Repeat the menu until the user selects Exit.
11. Stop.

C Program

```
#include <stdio.h>

void inputMatrix(int r, int c, int a[r][c]);
void printMatrix(int r, int c, int a[r][c]);
void additionDemo(void);
void multiplicationDemo(void);

void inputMatrix(int r, int c, int a[r][c]) {
    for (int i = 0; i < r; i++)
        for (int j = 0; j < c; j++)
            scanf("%d", &a[i][j]);
}

void printMatrix(int r, int c, int a[r][c]) {
```

```

    for (int i = 0; i < r; i++) {
        for (int j = 0; j < c; j++)
            printf("%d\t", a[i][j]);
        printf("\n");
    }
}

void additionDemo(void) {
    int r, c;
    printf("Enter rows and columns: ");
    scanf("%d %d", &r, &c);

    int A[r][c], B[r][c], AB[r][c], BA[r][c];
    printf("Enter matrix A:\n"); inputMatrix(r, c, A);
    printf("Enter matrix B:\n"); inputMatrix(r, c, B);

    for (int i = 0; i < r; i++)
        for (int j = 0; j < c; j++) {
            AB[i][j] = A[i][j] + B[i][j];
            BA[i][j] = B[i][j] + A[i][j];
        }

    printf("A + B:\n"); printMatrix(r, c, AB);
    printf("B + A:\n"); printMatrix(r, c, BA);
    printf("Since A + B = B + A, matrix addition is commutative.\n");
}

void multiplicationDemo(void) {
    int n;
    printf("Enter order of square matrices: ");
    scanf("%d", &n);

    int A[n][n], B[n][n], AB[n][n], BA[n][n];
    printf("Enter matrix A:\n"); inputMatrix(n, n, A);
    printf("Enter matrix B:\n"); inputMatrix(n, n, B);

    for (int i = 0; i < n; i++)
        for (int j = 0; j < n; j++) {
            AB[i][j] = BA[i][j] = 0;
            for (int k = 0; k < n; k++) {
                AB[i][j] += A[i][k] * B[k][j];
                BA[i][j] += B[i][k] * A[k][j];
            }
        }

    printf("A x B:\n"); printMatrix(n, n, AB);
    printf("B x A:\n"); printMatrix(n, n, BA);

    int same = 1;
    for (int i = 0; i < n; i++)
        for (int j = 0; j < n; j++)
            if (AB[i][j] != BA[i][j]) same = 0;

    if (!same)
        printf("A x B != B x A, so this proves matrix multiplication is not commutative.\n");
    else
        printf("These particular matrices give equal products. Try another pair; matrix multiplication is not commutative in general.\n");
}

int main(void) {
    int choice;
    do {

```

```

printf("\n1. Matrix Addition (Commutative)\n");
printf("2. Matrix Multiplication (Not Commutative)\n");
printf("3. Exit\nEnter choice: ");
scanf("%d", &choice);

switch (choice) {
    case 1: additionDemo(); break;
    case 2: multiplicationDemo(); break;
    case 3: printf("Exiting...\n"); break;
    default: printf("Invalid choice.\n");
}
} while (choice != 3);
return 0;
}

```

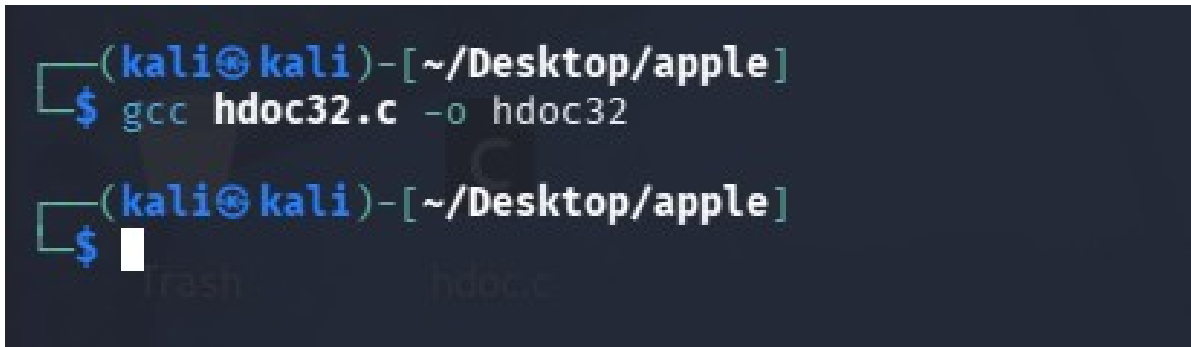
Suggested Test Values

Operation	Matrix A	Matrix B
Addition	1 2	5 6
	3 4	7 8
Multiplication	1 2	0 1
	3 4	1 0

Output Screenshots

Insert the terminal screenshots here after compiling and running the program.

Screenshot 1 - Compilation and main menu



```

(kali@kali)-[~/Desktop/apple]
$ gcc hdoc32.c -o hdoc32

(kali@kali)-[~/Desktop/apple]
$

```

Screenshot 2 - Addition result (A + B and B + A)

```

1. Matrix Addition (Commutative)
2. Matrix Multiplication (Not Commutative)
3. Exit
Enter choice: 1
Enter rows and columns: 2
2
Enter matrix A:
1 2 3 4
Enter matrix B:
5 6 7 8
A + B:
6      8
10     12
B + A:
6      8
10     12
Since A + B = B + A, matrix addition is commutative.

```

Screenshot 3 - Multiplication result (A x B and B x A)

```

1. Matrix Addition (Commutative)
2. Matrix Multiplication (Not Commutative)
3. Exit
Enter choice: 2
Enter order of square matrices: 2
Enter matrix A:
1 2 3 4
Enter matrix B:
0 1 1 0
A x B:
2      1
4      3
B x A:
3      4
1      2
A x B ≠ B x A, so this proves matrix multiplication is not commutative.

```

Screenshot 4 - Exit option

```

1. Matrix Addition (Commutative)
2. Matrix Multiplication (Not Commutative)
3. Exit
Enter choice: 3
Exiting...

```

```

(kali@kali)-[~/Desktop/apple]
$

```

Result

The menu-driven C program was implemented using separate functions. The output demonstrates that matrix addition is commutative ($A + B = B + A$), while matrix multiplication is not commutative in general ($A \times B \neq B \times A$ for the selected test matrices).